ORIGINAL ARTICLE

RFNBO Price Pathways and Renewable Hydrogen Plant Design: An Origin Aware Engineering Study of Selected European Market Areas

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ABSTRACT
Renewable hydrogen plants balance renewable capacity, storage, and grid electricity admitted by a price pathway. We develop an origin aware linear program that preserves electricity provenance in battery and hydrogen storage. Across 18 European cells, profile shifts change modeled savings by a median 0.82 percentage points and up to 4.86 points. The 2019 median saving is 1.31%; 16 of 18 cells are below 3%, while SK reaches 11.77% and EE has no response. Value therefore depends on temporal coincidence, not trigger frequency alone. This is an engineering screen, not legal certification or an EU 27 network estimate.

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1. Introduction

Renewable hydrogen projects must choose whether to meet a delivery target mainly by building renewable capacity and storage or by using permitted grid electricity during qualifying hours. That choice changes the optimal mix of wind, solar, electrolyzer capacity, storage, and imported electricity. The delegated regulation for renewable fuels of non biological origin (RFNBO) places temporal, geographical, and additionality conditions around electricity sourcing, while its price pathway gives a narrow route for treating grid electricity as renewable under specified conditions (European Commission 2023a,b). The engineering consequence is a design problem: under an explicit rule proxy, when can qualifying grid imports replace costly renewable overbuild or storage, and when does the least cost design remain unchanged?

The central engineering uncertainty is whether eligible price hours arrive when the renewable only design is short of renewable supply; a high trigger count alone need not create a capacity margin.

Existing European studies establish why the question matters. Brandt et al. assess the cost and competitiveness consequences of the European regulatory framework (Brandt et al. 2024). Ruhnau and Schiele evaluate how relaxing simultaneity changes hydrogen project design, economics, and power sector emissions (Ruhnau and Schiele 2023). Zeyen et al. examine temporal regulation of renewable supply, while Namazifard et al. compare unrestricted and green electrolytic hydrogen under EU criteria (Zeyen,

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Riepin, and Brown 2024; Namazifard et al. 2025). Studies of portfolio effects, PPAs, carbon accounting incentives, and spatial design extend the discussion to system, contractual, or European wide settings (Casas Ferrus, Ruhnau, and Madlener 2024; Langer et al. 2026; Linsel and Bertsch 2026; Radek, Breder, and Weber 2026; Giovanniello et al. 2024; Glenk, Holler, and Reichelstein 2026; Li, Beck, and Thomas 2026; Marocco, Gandiglio, and Santarelli 2026). Complementary project studies analyze curtailment, active trading, renewable configurations, and temporal correlation (Ganter et al. 2025; Georgopoulos et al. 2025; Adekola et al. 2024; Sgaramella et al. 2023; Pinheiro et al. 2025). Recent preprints also address cost upper bounds, PPA uncertainty, exporters, and market participation (Hordvei et al. 2024; Brandt, Bensmann, and Hanke-Rauschenbach 2026; Schumm et al. 2024; Johnsen et al. 2025).

Three features of that literature sharpen the present question. First, system wide studies appropriately optimize electricity sector interactions or emissions but do not necessarily identify the project boundary capacity substitutions that a price pathway proxy enables. Second, studies that relax temporal matching show that timing matters, yet a comparison of relaxed rules alone does not reveal whether the frequency of eligible price hours or their alignment with a plant's renewable profile creates the economic margin. Third, models that combine electricity and hydrogen storage can produce different designs depending on whether energy provenance is retained after charging. For a rule sensitive plant sizing exercise, allowing a grid origin charge to inherit renewable provenance would change the accounting problem itself. The practical gap is therefore not another map of average hydrogen cost: it is a transparent test of which timing and accounting conditions turn an expanded rule pathway into a material design difference.

Existing studies establish that timing and regulatory design matter for hydrogen costs and emissions (Brandt et al. 2024; Ruhnau and Schiele 2023; Zeyen, Riepin, and Brown 2024; Namazifard et al. 2025). Three engineering questions remain. First, how large is the design response when the price pathway is isolated from other regulatory elements? Second, is the response driven by eligible hour frequency or by temporal coincidence between low price hours and renewable output? Third, does the response transfer across years, public profiles, and bounded technology assumptions? These are plant design mechanism questions rather than system cost questions. They require origin preserving storage accounting and a temporal phase shift experiment, which we implement in a common linear programming framework.

This study therefore asks three engineering questions. **RQ1:** Under an origin aware price pathway proxy, how large are the LCOH and capacity differences from the renewable only baseline in a selected public data screen? **RQ2:** Holding price traces, EUA values, trigger frequency, delivery target, and engineering caps fixed, does changing only the timing of wind and solar profiles materially change those differences? **RQ3:** Do the magnitude, branch attribution, and capacity response transfer across retained years, public published day ahead profiles, and bounded technology cost assumptions?

The paper's differentiating contribution is an auditable, reproducible, conditional engineering comparison. It integrates origin tagged storage accounting and price branch decomposition in a single linear program, proves the nested feasible set property that separates structural direction from empirical magnitude, and evaluates a predeclared temporal shift mechanism, cross period/profile sensitivity, and finite parameter stress. The estimand is the difference in least cost model boundary LCOH and asset capacity for selected, separately solved public data market area/profile cells. This design makes the timing mechanism and its capacity consequences inspectable without treating a local response as a European wide property.

The scope is fixed at the estimand level. This is a public data engineering screening study. The estimand is the conditional difference in least cost model boundary LCOH and asset capacity for selected, separately solved public market area/profile cells. Legal RFNBO certification, including complete additionality, geographical correlation, contracting, metering, support, and production evidence, is outside the estimand. The study also does not estimate EU-27 project level prevalence, project level congestion, bidding zone congestion, point forecasts, authenticated forecast vintage performance, real time operation, or site specific interconnection, metering, and contract outcomes. Within this boundary, the model preserves renewable and grid provenance in separate storage ledgers and uses controlled timing, cross period, and bounded parameter comparisons to identify conditional design responses.

2. Materials and methods

2.1. *Study boundary and research design*

The study is a selected European public data engineering screen. The primary layer has 18 complete 2019 cells, each pairing a public wind/solar generation shape with a public day ahead price area. DE-LU is represented as its coupled market area; DK1 and IT North are paired with country level renewable profile shapes. These cells were retained for complete paired hourly data and profile diversity, not sampled probabilistically from projects or EU bidding zones. We selected 2019 as the primary screen because it provides a before the COVID 19 pandemic historical benchmark with the common paired hourly coverage retained across all 18 cells; E2 tests transfer to later years and alternative published profiles. Each cell has a fixed annual hydrogen delivery target, scaled to retained UTC hours, a 250 MW source line proxy, and a 1,000 MW grid import proxy. The model compares a renewable only baseline with declared computational price pathway cases at the same target and cost boundary.

The analysis is interpreted within this declared study boundary. The 2019 hourly traces serve as a historical engineering stress test of a 2030 rule representation and are used comparatively rather than as 2030 price forecasts.

Table 1 summarizes the predeclared evidence sequence.

2.2. *Rule proxy and nested feasible sets*

The model implements a computational price pathway proxy. In hour t, the low price indicator is

$$I_t = \mathbb{1}\{p_t \leq 20 \text{ EUR/MWh } \vee \ p_t < 0.36E_t\},$$

where p_t is the day ahead price and E_t is the daily EUA auction price carried across the corresponding UTC day. The implementation separately solves negative price only, EUR 20/MWh only, EUA only, joint, and renewable only cases when the experiment calls for branch decomposition. The EUR 20/MWh threshold represents the low price branch encoded by the RFNBO rule proxy, whereas $0.36E_t$ is the declared computational signal used for the EUA linked branch. The latter is not treated as legal equivalence. Hereafter, "price pathway proxy" denotes the rule representation, and "model qualified grid electricity" denotes grid energy admitted by that proxy and selected by the optimization; neither phrase is a legal certification. The indicator is a transparent computational signal, while legal certification and site evidence questions remain outside the estimand. This separation makes the price logic explicit so it can be stress tested reproducibly.

The direction of the price pathway comparison has a simple structural property. Let $\mathcal{F}_R$ be the renewable only feasible set and $\mathcal{F}_P$ the price pathway feasible set. A renewable only solution remains feasible under the price pathway formulation by setting all model qualified grid flows and all grid origin storage flows to zero. All remaining balances and the delivery target are unchanged. Thus $\mathcal{F}_R \subseteq \mathcal{F}_P$. With the same objective, the optimized costs obey $J_P \leq J_R$. Nonnegative modeled savings therefore follow from the formulation.

What the model identifies. Because the feasible sets are nested, cost direction is structural rather than empirical. The empirical contribution begins with three quantities that are not structural: (i) the magnitude of strict savings, which indicates whether the pathway is materially useful in a cell; (ii) the capacity substitution pattern, which identifies the renewable or conversion assets the optimizer avoids; and (iii) the response to controlled timing shifts, which tests how temporal coincidence changes the substitution margin. Each quantity is conditional on the public profile, price trace, delivery target, and declared cost box. This is the basis for interpreting the results below.

2.3. *Origin aware linear program*

For each cell, the linear program chooses wind capacity K_W, solar capacity K_S, battery power and energy (K_B, E_B), electrolyzer capacity K_E, hydrogen storage K_H, and hourly electricity/hydrogen flows. It minimizes annualized capacity cost plus variable electricity, grid fee, and battery degradation cost:

$$\min_{K,x} J = \sum_i a_i c_i K_i + \sum_t \Delta t \left[(p_t + f_g) g_t + c_B b_t \right] ,$$

where $a_i c_i$ is the annualized unit cost of asset i, g_t is grid import, f_g is the declared grid fee, and b_t is battery throughput. Screening LCOH is J divided by the fixed delivered hydrogen quantity. The objective includes wind, solar, battery, electrolyzer, hydrogen storage, declared electricity, grid fee, and degradation terms. It excludes water, compression, transport, taxes, financing, lifecycle inventories, project development, and offtake costs.

Under the central cost vector, the optimizer selects zero battery capacity in all 18 primary price pathway solutions. We do not impose this outcome. It follows from the joint interaction of annualized asset costs, conversion efficiencies, storage losses, and hourly operating costs; E3 tests how the resulting cost and capacity response changes within the bounded parameter box.

Renewable availability is bounded by $K_W CF_t^W + K_S CF_t^S$, and source line and grid import limits constrain hourly flows. Renewable- and grid origin battery inventories are separate. Renewable origin and non renewable hydrogen inventories are also separate. In the model qualified route, electrolyzer load credited by the model is balanced only by renewable direct flow, renewable tagged battery discharge, and qualifying grid input. Terminal storage states are zero, which prevents endpoint energy from reducing the objective artificially. A cumulative renewable energy safeguard restricts model qualified grid energy to no more than optimized renewable generation over the represented horizon. This is a conservative quantity surrogate, not a complete Article 5 (RFNBO additionality) legal test.

This accounting structure makes the comparison interpretable. Direct renewable electricity, renewable electricity discharged from the battery, and model qualified grid electricity are mutually identified in every represented hour. Grid origin battery energy remains in its own ledger and cannot support model qualified hydrogen delivery. The annual quantity safeguard is applied after hourly balances rather than substituted for them; it cannot create qualifying electricity in an hour that fails the price indicator. Conversely, a low price hour does not force grid import: the optimizer uses it only if it reduces the declared total cost while meeting the same delivery target and all source line limits. This distinction matters for interpreting a capacity response. A reduction in solar or electrolyzer capacity is not assigned mechanically by the rule; it is an optimized substitution that occurs only when the timing, price, and capacity constraints make it less costly than renewable overbuild.

The complete code aligned balance equations, bounds, and variable definitions are supplied in the reproducibility package. The main text presents the governing objective and accounting logic because the storage provenance control is central: a grid import cannot retrospectively turn renewable storage into grid origin energy, and a renewable charge cannot be counted twice.

2.4. *Public data, processing, and reproducibility*

The primary 2019 generation profiles use the Energy-Charts public API and country generation series (Energy-Charts 2026a,b). Primary day ahead price series use the public archived ENTSO-E-derived snapshot with its recorded dataset DOI (De Felice 2020). Cross period DE-LU price and generation series use publicly accessible Energy-Charts/SMARD sources, while Belgian observed and published day ahead profile series use Elia public data (Bundesnetzagentur 2026; Elia 2026). EUA prices are drawn from public EEX market data (European Energy Exchange 2026). Technology assumptions are declared from the Danish Energy Agency technology data source and fixed engineering parameters (Danish Energy Agency 2026). Area identifiers and access conventions follow ENTSO-E public documentation (ENTSO-E 2026b,a).

The public data requirement is a design constraint rather than a rhetorical feature. The analysis uses no private plant dispatch, confidential connection agreement, proprietary forecast archive, or paid market database. Each derived cache records a source path, UTC coverage, transformation manifest, and SHA-256 digest. A source is retained only when the paired hourly fields required by its experiment are available under the documented public access route. This produces a narrower estimand than a project study, but it permits an independent researcher to inspect the inputs and rerun the comparison without a commercial data license or an operator partnership.

All model timestamps are UTC. Retained primary cells contain 8,687 to 8,760 paired hours, and no missing hour is imputed. E2 retains every planned source year profile with at least 99% complete hours; the achieved minimum was 99.726%. Local calendars select years only; daylight saving transitions are represented through the underlying UTC hour sequence. The Belgian published day ahead profiles do not carry an authenticated issue time field in the retained public source. They are therefore a public published profile sensitivity only, not forecast vintage validation, real time evidence, or a forecast error estimate.

The base cost vector is fixed before the experiments: 120,000 EUR/MW/year for wind, 85,000 EUR/MW/year for solar, 45,000 EUR/MW/year for battery power, 12,000 EUR/MWh/year for battery energy, 100,000 EUR/MW/year for the electrolyzer, and 2 EUR/kg/year for hydrogen storage. Renewable variable cost is 1 EUR/MWh, the base grid fee is 3 EUR/MWh, and battery degradation is 2 EUR/MWh. The base hydrogen yield is 20 kg/MWh; charge and discharge efficiencies are 0.90 for the battery and 0.995 for hydrogen. These are annualized engineering assumptions, not observed project quotations. E3 varies the declared grid fee and other bounded assumptions around this same reference configuration.

The derived public data cache, scenario manifest, solver checks, numerical claim ledger, code hash inventory, and editable result tables were stored in the reproducibility package used for this manuscript. The package contains the transformation record and the public source identifiers needed to inspect the reported calculations; no authenticated credential or private dataset is used.

2.5. *Experiments and technical checks*

E1: temporal phase shift. For every primary cell, wind and solar curves were jointly circularly shifted by 0, 2,190, 4,380, and 6,570 retained hourly positions. Prices, EUA values, rule trigger counts, delivery target, costs, and engineering caps were fixed. This is a synthetic timing perturbation, not a realized operating case. It isolates temporal coincidence while keeping each cell's price distribution and trigger frequency unchanged.

E2: cross period and public published profile sensitivity. We solved five cases (renewable only, negative only, EUR 20/MWh only, EUA only, and joint) for DE-LU observed 2019-2022; BE observed 2021-2025; and BE published day ahead profiles for 2021-2025. All source year profiles meeting the prespecified completeness criterion were retained. The comparison asks whether branch attribution and magnitude change across public hourly curves, rather than treating one historical year as general.

E3: bounded parameter stress. The 18 primary cells were run under the base parameterization plus 12 fixed seed Latin hypercube draws. The finite stress box varies CAPEX by 0.80-1.20, hydrogen yield by 0.90-1.10, grid fee from 0-6 EUR/MWh, battery charge/discharge efficiency from 0.85-0.95, hydrogen charge/discharge efficiency from 0.98-1.00, and the EUA proxy by 0.80-1.20. These are engineering ranges, not distributions. Consequently, ranges reported below are neither confidence intervals nor probability statements.

The audit required every solver status to equal zero, delivery target error below 1e-4 kg, electricity/source line/eligible grid excess below 1e-5 MW, zero terminal storage states, strictly UTC timestamps, valid cache hashes, and complete planned scenario coverage. The phase shift zero position reproduced the pre upgrade primary baseline and joint LCOH values to a maximum absolute error of 0.0000000005 EUR/kg. A separate 48-hour extreme target control was infeasible as expected and is retained as a solver boundary check in the supporting materials.

Verification and validation are kept distinct. Verification concerns whether the declared optimization and data transformations execute as specified; the balance, target, cap, UTC, cache, zero shift, and infeasibility controls address that question. Project level validation would require publicly reproducible measurements of a plant's connection, renewable contract, installed assets, operational dispatch, and realized costs. Those inputs are unavailable for the selected European screen and are not imputed. Comparisons with the literature therefore serve as mechanism triangulation, not numerical calibration: the study tests a transparent conditional capacity and cost response rather than claiming to reproduce a particular project's LCOH.

3. Results

3.1. *Temporal coincidence as the mechanism (E1)*

E1 keeps the price trace, EUA proxy, trigger set, delivery target, costs, and capacity limits fixed while jointly rotating the wind and solar profiles by 0, 2,190, 4,380, and 6,570 retained hourly positions. Across cells, the within cell price pathway saving range has a median of 0.82 percentage points and a maximum of 4.86 points in SK. 16 of 18 cells have a saving range of at least 0.25 percentage points. The corresponding median range in the renewable price overlap indicator is 0.654.

The phase shift design gives a direct within model mechanism test. Price distributions and trigger counts are invariant by construction, yet changing when wind and solar occur changes both LCOH saving and capacity substitutions. The descriptive within cell slope between the prespecified overlap indicator and saving is -2.30 percentage points per unit overlap ratio. On this descriptive scale, a 0.10 change in the overlap ratio corresponds to 0.23 percentage points in modeled saving; this is not an inferential effect estimate. Its negative sign is consistent with the engineering intuition that greater renewable output during low price windows leaves less scope for qualifying grid energy to replace renewable overbuild. Figure 1 reports all cell level shifts and the associated capacity response range, preserving the full set of timing positions.

E1 is a synthetic timing perturbation that preserves each marginal series while changing their chronology. The observed within cell ranges show that the joint comparison depends on the alignment between a fixed price chronology and renewable availability. Within the declared plant design problem, the incremental value of a qualifying import hour depends on two factors: the renewable supply that would otherwise have been available in that hour, and the capacity required to serve the unchanged target. This identifies temporal coincidence as the operative mechanism in the model configuration.

3.2. *Magnitude and capacity substitution (primary screen)*

The primary comparison contains 17 cells with a strictly positive modeled saving and 1 zero response cell. This direction count is a consequence of the nested feasible sets; the informative result is the size of strict improvement and its associated capacity response. The median joint pathway saving is 1.31%, and the maximum is 11.77% in SK. Figure 2 displays LCOH and capacity differences across the 18 cells.

SK illustrates an economically material capacity response under the declared boundary. Its price pathway solution reduces solar capacity by 58.8 MW and electrolyzer capacity by 20.6 MW relative to the renewable only design while using 74,547 MWh of model qualified grid electricity. The LCOH difference is 0.621 EUR/kg. At zero shift, the computational price indicator flags 436 hours in SK and 295 hours in EE; the corresponding renewable price overlap ratios are 0.922 and 0.938, respectively. EE is a useful zero response boundary: its overlap ratio is slightly higher than SK's, yet its model qualified grid use is 0 MWh and its optimized capacities remain unchanged. This pair shows that trigger frequency or overlap ratio alone do not suffice to predict response. The active substitution margin also depends on renewable only capacity scarcity and the relative cost of substitution. Across the central screen, the optimizer chooses zero battery capacity in these cells.

Capacity differences are heterogeneous, so their medians are descriptive summaries of the selected cells. Across the 18 cells, the median joint minus baseline changes are -2.1 MW for wind, -0.3 MW for solar, and 0.3 MW for electrolyzer capacity. The count of price pathway solutions with nonzero battery capacity is 0 of 18 under this declared central cost box. The central screen therefore identifies a renewable overbuild and capacity timing margin; it does not establish a general technology ranking for batteries or other RFNBO project designs.

3.3. *Cross period, branch, and bounded parameter sensitivity (E2 and E3)*

The mechanism identified in Section 3.1 is tested against changes in year, public profile, price branch, and bounded engineering assumptions. E2 completed 70 optimizations across 14 complete source year/profile cells. Joint savings range from 0.90% to 18.78%, with a median of 3.44%. The expanded record shows why the 2019 branch attribution is cell specific. In 2019 DE-LU, the joint and EUR 20/MWh only cases coincide to the model's displayed precision (joint minus EUR 20/MWh only LCOH = 0.00000000 EUR/kg). Across the retained cross period screen, however, the largest absolute joint minus EUR-20 difference is 0.0328 EUR/kg. The EUA linked component is small in the observed expanded cells but remains active in some cells.

The five paired Belgian public profiles provide an input curve sensitivity. In every paired year, the published day ahead profile produces a lower saving than the observed profile by 0.13 to 1.12 percentage points. The comparison shows that switching the renewable curve alone can alter the engineering result, consistent with E1's temporal coincidence mechanism. These published profiles are treated as a sensitivity input, not as authenticated forecast vintage validation.

The E2 protocol reports all 14 source year/profile cells that met the stated 99% completeness threshold. The spread across DE-LU 2019-2022 and Belgian 2021-2025 curves is evidence about the transfer conditions of the screen, rather than a count of independent project observations. Together with E1, E2 supports a precise engineering statement: the shape and timing of the public input trace affect both the size of the substitution margin and the apparent contribution of individual price branches.

E3 completed 468 optimizations: the base case plus 12 fixed seed bounded draws for each primary cell and each of two cases. Across this finite engineering box, cell level saving values range from 0.00% to 12.06%. The largest within cell span is 0.86 percentage points in SK. There are 12 zero response draws and 222 positive response draws. These results map response boundaries inside the declared engineering box and retain both zero and positive responses.

Figure 4 reports the full minimum to maximum range for every cell. The stress design varies annualized asset costs, conversion yield, grid fee, storage losses, and the EUA price proxy jointly, so the range tests whether the central response depends on one calibration. It is a bounded condition map; the draws have no attached likelihood, and the reported ranges are not confidence intervals. Capacity upper bound hits and all feasibility checks remain in the audit.

BE and CZ provide intermediate stress cases rather than endpoints. Their finite saving ranges are 3.20% to 3.64% and 1.66% to 1.90%, respectively. Including these cells shows that the central interpretation is not based only on the largest response and the zero response boundary.

4. Discussion

4.1. *The central mechanism: temporal coincidence, not trigger frequency*

The central engineering finding is that temporal coincidence is a primary determinant of the substitution margin within the declared model. E1 holds the low price pattern, EUA proxy, trigger count, delivery target, cost vector, and capacity limits fixed while changing only renewable chronology. The resulting median 0.82-percentage point within cell range, reaching 4.86 points, shows that the value of qualifying grid electricity depends on when renewable output would otherwise be available. Trigger frequency is therefore an incomplete design diagnostic.

A price rule creates design value when its qualifying hours coincide with a scarcity margin in the renewable only design. The selected cells illustrate the endpoints: SK reaches 11.77% saving with reductions of 58.8 MW in solar capacity and 20.6 MW in electrolyzer capacity, whereas EE has zero modeled response under the same declared cost structure. E1 supplies the controlled within cell evidence behind this contrast. Reporting average electricity prices or trigger counts without their alignment with renewable output can miss the mechanism.

4.2. *Interpretation boundary*

The declared estimand keeps the engineering result proportionate to its evidence. The paper evaluates conditional least cost differences for selected public market area/profile cells under a transparent price proxy, source preserving storage accounting, and declared capacity caps. Its value is the inspectable response surface: readers can see which input chronology and capacity margin produce a change, and which produce zero response. Project eligibility, site evidence, network physics, and broader European prevalence require a separate evidence layer.

4.3. *Relationship to previous work*

The study complements broader European regulatory and system analyses. Brandt et al. quantify the cost and competitiveness consequences of the European regulatory framework; Ruhnau and Schiele study how simultaneity affects project design, economics, and power sector emissions; Zeyen et al. analyze temporal regulation; and Namazifard et al. compare unrestricted and green electrolytic hydrogen under EU criteria (Brandt et al. 2024; Ruhnau and Schiele 2023; Zeyen, Riepin, and Brown 2024; Namazifard et al. 2025). PPA, additionality, allocation, and carbon accounting studies address system or contract design consequences (Casas Ferrus, Ruhnau, and Madlener 2024; Langer et al. 2026; Linsel and Bertsch 2026; Radek, Breder, and Weber 2026; Giovanniello et al. 2024; Glenk, Holler, and Reichelstein 2026; Li, Beck, and Thomas 2026). Project configuration studies show why renewable composition, curtailment, active trading, and temporal correlation matter (Ganter et al. 2025; Georgopoulos et al. 2025; Adekola et al. 2024; Sgaramella et al. 2023; Pinheiro et al. 2025).

The present study adds a narrow but testable capability. It combines source preserving storage accounting and price branch decomposition with a controlled phase shift experiment that preserves price statistics. The nesting proof separates structural cost direction from empirical magnitude, while E1 identifies the timing condition behind the substitution margin. E2 further shows that the 2019 joint and EUR 20/MWh only cases coincide at displayed precision in DE-LU, whereas the expanded record contains a maximum absolute difference of 0.0328 EUR/kg. The result is a conditional mechanism benchmark that complements system wide and project specific studies.

4.4. *Implications for policy and practice*

For policy assessment. A price rule assessment should report the qualifying hour set, renewable price overlap, and capacity substitutions alongside LCOH. A single European cost multiplier can conceal the condition that creates value: low price eligibility arriving when on site renewable output is comparatively scarce. The evidence supports differentiated screening by market area and year, while leaving the regulatory threshold itself to formal legal and policy analysis.

For project screening. The model offers a sequence of engineering checks. A promising cell should be followed by project specific verification of interconnection rating, meterable renewable supply, contract evidence, geographical relationship, and the complete regulatory record. A zero response cell is also informative because it shows that price eligibility alone need not change the least cost design. The bounded parameter results support reporting response ranges and zero response cases alongside a central LCOH.

4.5. *Limitations and validation boundary*

The limitations are paired with deliberate mitigations. First, the historical traces and perfect foresight define a comparative static design screen; E1 and E2 test transfer across chronology and public years, while operational control remains a separate question. Second, country profile/market area pairs are transparent proxies; the complete source rule and the paired Belgian profile comparison make that substitution visible. Third, constant source line and import caps are engineering stress proxies; the manuscript treats their responses as capacity boundary evidence and reserves flow based network analysis for a project level extension. Fourth, the price indicator and cumulative energy safeguard define a computational proxy, so legal certification remains outside the estimand. Fifth, E3 reports a finite engineering range; the full draw set is retained without probability or confidence interpretation. Finally, the Belgian published day ahead profiles provide an input sensitivity rather than authenticated forecast vintage evidence.

These choices make the current estimand reproducible and identify the next evidence rich extension: a documented interconnection location, authenticated market and forecast vintages, complete contractual and metering records, and network constraints. The present contribution is the intermediate engineering layer that shows where those project level data would change the design conclusion.

5. Conclusions

The controlled phase shift experiment shows that timing changes the value of a price pathway proxy. With prices, EUA values, the eligible hour set, delivery target, costs, and caps fixed, rotating the renewable profiles changes modeled savings by a median 0.82 percentage points and up to 4.86 points. The nested feasible set proof means that the nonnegative direction is structural; the magnitude, capacity substitutions, and timing response are the engineering evidence.

For engineering screening, average price or trigger frequency alone is insufficient. The primary 2019 screen has a median saving of 1.31%; 16 of 18 cells are below 3%, while SK reaches 11.77% and EE has no modeled response. Screening should report the eligible hour set, the renewable price overlap ratio, and capacity substitutions together. Under the central cost vector, the primary price pathway solutions use no battery capacity, so storage choices remain conditional on the declared cost and efficiency assumptions.

The evidence remains conditional on the selected public profiles, perfect foresight, computational rule proxy, and capacity caps. E2 shows that the magnitude and branch attribution can change across retained years and public input curves; E3 maps finite response bounds without probability or confidence interpretation. The next extension should add rolling or imperfect foresight, documented site interconnection and flow based network constraints, and project evidence needed for legal certification. This study does not estimate EU 27 prevalence, congestion, bankability, or compliance.

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CRediT authorship contribution statement

Junjie Zhang: Conceptualization, Methodology, Software, Data curation, Formal analysis, Visualization, Investigation, Writing - original draft, Writing - review and editing.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Generative AI use statement

Generative AI tools were used only to assist Python code development and English language editing. The author reviewed and verified the analysis, references, interpretation, and final manuscript, and took responsibility for the reported work.

Data availability

All source datasets are publicly accessible from the cited providers. The reproducibility package accompanying this manuscript contains the derived cache manifest, scenario definitions, solver audit, result tables, figures, and code. Raw provider files are not redistributed. Provider links, identifiers, processing steps, and cache hashes are recorded in the package. The versioned package used for this manuscript is available at `https://github.com/williamjay1/rfnbo-hydrogen-europe-ijge/releases/tag/v1.0.1`.

References

Adekola, O., E. Ghafoori, B. Dechamp, and A. Prada. 2024. "Electricity supply configurations for green hydrogen hubs: A European case study on decarbonizing urban transport." *International Journal of Hydrogen Energy* 85: 539–559. https://doi.org/10.1016/j.ijhydene.2024.08.336.

Brandt, J., A. Bensmann, and R. Hanke-Rauschenbach. 2026. "Mitigating business risks from renewable PPA power sourcing uncertainties for European green hydrogen production: Robust system design, regulatory adjustments and offtake flexibility." `https://arxiv.org/abs/2606.13215`.

Brandt, J., et al. 2024. "Cost and competitiveness of green hydrogen and the effects of the European Union regulatory framework." *Nature Energy* 9: 703–713. https://doi.org/10.1038/s41560-024-01511-z.

Bundesnetzagentur. 2026. "SMARD electricity market data." `https://www.smard.de/en`.

Casas Ferrus, M. N., O. Ruhnau, and R. Madlener. 2024. "Portfolio effects in green hydrogen production under temporal matching requirements." *Energy Strategy Reviews* 56: 101580. https://doi.org/10.1016/j.esr.2024.101580.

Danish Energy Agency. 2026. "Technology data for renewable fuels and energy technologies." `https://ens.dk/en/analyses-and-statistics/technology-data-renewable-fuels`.

De Felice, M. 2020. “entsoe-hourly-prices.csv.” Figshare dataset. CC BY 4.0, https://figshare.com/articles/dataset/entsoe-hourly-prices_csv/12178929.
Elia. 2026. “Power generation data.” https://www.elia.be/en/grid-data/power-generation.
Energy-Charts. 2026a. “Public electricity market and generation data.” https://www.energy-charts.info/.
Energy-Charts. 2026b. “Public power API documentation and country generation series.” https://api.energy-charts.info/.
ENTSO-E. 2026a. “Area list with Energy Identification Code (EIC).” https://transparencyplatform.zendesk.com/hc/en-us/articles/15885757676308-Area-List-with-Energy-Identification-Code-EIC.
ENTSO-E. 2026b. “How to get a security token and Transparency Platform API request methods.” https://transparencyplatform.zendesk.com/hc/en-us/articles/12845911031188-How-to-get-security-token.
European Commission. 2023a. “Commission Delegated Regulation (EU) 2023/1184 supplementing Directive (EU) 2018/2001.” EUR-Lex. https://eur-lex.europa.eu/eli/reg_del/2023/1184/oj.
European Commission. 2023b. “Commission Delegated Regulation (EU) 2023/1185 supplementing Directive (EU) 2018/2001.” EUR-Lex. https://eur-lex.europa.eu/eli/reg_del/2023/1185/oj.
European Energy Exchange. 2026. “Market data downloads and environmental products.” https://www.eex.com/en/downloads-3.
Ganter, R., T. H. Ruggles, P. Gabrielli, G. Sansavini, and K. Caldeira. 2025. “Utilizing curtailed wind and solar power to scale up electrolytic hydrogen production in Europe.” *Environmental Science & Technology* 59 (7): 3495–3507. https://doi.org/10.1021/acs.est.4c10168.
Georgopoulos, A., A. Papadopoulos, P. Mitkidis, and E. Giannissi. 2025. “Active trading and regulatory incentives lower the levelized cost of green hydrogen in Greece.” *Communications Earth & Environment* 6 (1): 370. https://doi.org/10.1038/s43247-025-02349-3.
Giovanniello, M., A. Cybulsky, T. Schittekatte, and D. S. Mallapragada. 2024. “The influence of additionality and time-matching requirements on the emissions from grid-connected hydrogen production.” *Nature Energy* 9 (2): 197–207. https://doi.org/10.1038/s41560-023-01435-0.
Glenk, G., P. Holler, and S. Reichelstein. 2026. “How carbon accounting rules shape incentives for hydrogen production.” *Nature Communications* 17 (1): 7260. https://doi.org/10.1038/s41467-026-75473-z.
Hordvei, E., S. E. Hummelen, M. Petersen, S. Backe, and P. Crespo del Granado. 2024. “From policy to practice: Upper bound cost estimates of Europe’s green hydrogen ambitions.” https://arxiv.org/abs/2406.07149.
Johnsen, A. G., L. Mitridati, J. Kazempour, and L. Roald. 2025. “Electrolyzers bidding in electricity markets under green hydrogen regulations and uncertainty.” https://arxiv.org/abs/2507.20702.
Langer, L., A. Bjørn, B. F. Hobbs, M. Brander, and R. Bramstoft. 2026. “Renewable hydrogen: The system-wide impact of PPAs and additionality requirements for selected European countries.” *Energy Policy* 215: 115295. https://doi.org/10.1016/j.enpol.2026.115295.
Li, Y., F. Beck, and D. Thomas. 2026. “How the interplay between grid characteristics and renewable resources determines optimal temporal correlation for grid-connected hydrogen certification.” *Energy Policy* 212: 115158. https://doi.org/10.1016/j.enpol.2026.115158.
Linsel, O., and V. Bertsch. 2026. “Effects of the delegated act on renewable fuels of non-biological origin on production costs and distribution of hydrogen production capacities across Europe.” *Energy Policy* 212: 115178. https://doi.org/10.1016/j.enpol.2026.115178.
Marocco, P., M. Gandiglio, and M. Santarelli. 2026. “Optimising green hydrogen production across Europe: How renewable energy sources shape plant design and costs.” *Renewable Energy* 256: 124542. https://doi.org/10.1016/j.renene.2025.124542.
Namazifard, N., M. A. Tahavori, W. Nijs, P. Vingerhoets, and E. Delarue. 2025. “Relaxing EU hydrogen criteria: A cost and emission comparison of unrestricted and green electrolytic hydrogen in 2030.” *Environmental Research: Energy* 2: 035014. https://doi.org/10.1088/2753-3751/ae07e3.
Pinheiro, J., M. Gomes, F. Tofoli, L. Sampaio, F. Melo, J. Gregory, E. Sgro, and R. Leao. 2025. “Techno-economic analysis of green hydrogen generation from combined wind and photovoltaic systems based on hourly temporal correlation.” *International Journal of Hydrogen Energy* 97: 690–707. https://doi.org/10.1016/j.ijhydene.2024.11.429.
Radek, M., M. S. Breder, and C. Weber. 2026. “Hydrogen in the European power sector – A case

study on the impacts of regulatory frameworks for green hydrogen.” *Energy Policy* 213: 115200. https://doi.org/10.1016/j.enpol.2026.115200.

Ruhnau, O., and J. Schiele. 2023. “Flexible green hydrogen: The effect of relaxing simultaneity requirements on project design, economics, and power sector emissions.” *Energy Policy* 182: 113763. https://doi.org/10.1016/j.enpol.2023.113763.

Schumm, L., H. Abdel-Khalek, T. Brown, F. Ueckerdt, M. Sterner, D. Fioriti, and M. Parzen. 2024. “The impact of temporal hydrogen regulation on hydrogen exporters and their domestic energy transition.” `https://arxiv.org/abs/2405.14717`.

Sgaramella, G., L. M. Pastore, G. Lo Basso, and L. de Santoli. 2023. “Optimal RES integration for matching the Italian hydrogen strategy requirements.” *Renewable Energy* 219: 119409. https://doi.org/10.1016/j.renene.2023.119409.

Zeyen, E., I. Riepin, and T. Brown. 2024. “Temporal regulation of renewable supply for electrolytic hydrogen.” *Environmental Research Letters* 19: 024034. https://doi.org/10.1088/1748-9326/ad2239.

Table 1. Study design and evidence layers.

Experiment	Design	Fixed or varied components	Primary readout
Primary public data screen	18 complete 2019 cells; hourly origin aware least cost comparison	Price, renewable profiles, fixed delivery target, cost assumptions and engineering caps	LCOH and capacity differences between nested feasible sets
E1 temporal phase shift	Four circular shifts (0, 2,190, 4,380, and 6,570 h) for each cell	Prices, EUA proxy, trigger count, target and caps fixed; wind and solar timing varied	Within cell LCOH saving and capacity response range
E2 cross period/profile sensitivity	DE-LU observed 2019-2022; BE observed and published day ahead profiles 2021-2025	Hourly profile and year varied; five trigger cases compared	Magnitude, branch differences, and published profile sensitivity
E3 bounded engineering stress	Base case plus 12 fixed seed Latin hypercube draws in each primary cell	CAPEX, yield, grid fee, battery/hydrogen efficiencies and EUA factor varied within declared bounds	Finite response and capacity ranges; no probabilistic interpretation

Table 2. Public inputs and declared engineering assumptions.

Input	Source and access	Resolution and processing	Function and boundary
Wind and solar shapes	Energy-Charts public API	Hourly UTC, paired complete cells, no imputation	Country profile proxy; not plant SCADA
Primary prices	Public Figshare ENTSO-E-derived archive	Hourly UTC, 2019	Bidding area price proxy
Cross period profiles/prices	Energy-Charts/SMARD and Elia public series	Hourly UTC, retained profiles >=99% complete	E2 sensitivity; Belgian day ahead profiles lack issue time proof
EUA proxy	EEX primary auction reports	Daily auction value mapped to UTC hours	Price branch proxy only
Costs and efficiencies	Danish Energy Agency source plus stated vector	Annualized base values and bounded E3 perturbations	Engineering screen; not project quotations
Source line and import caps	Declared 250 MW and 1,000 MW constants	Applied hourly	Capacity stress proxy; not flow based network physics

Table 3. Primary 2019 selected cell comparison.

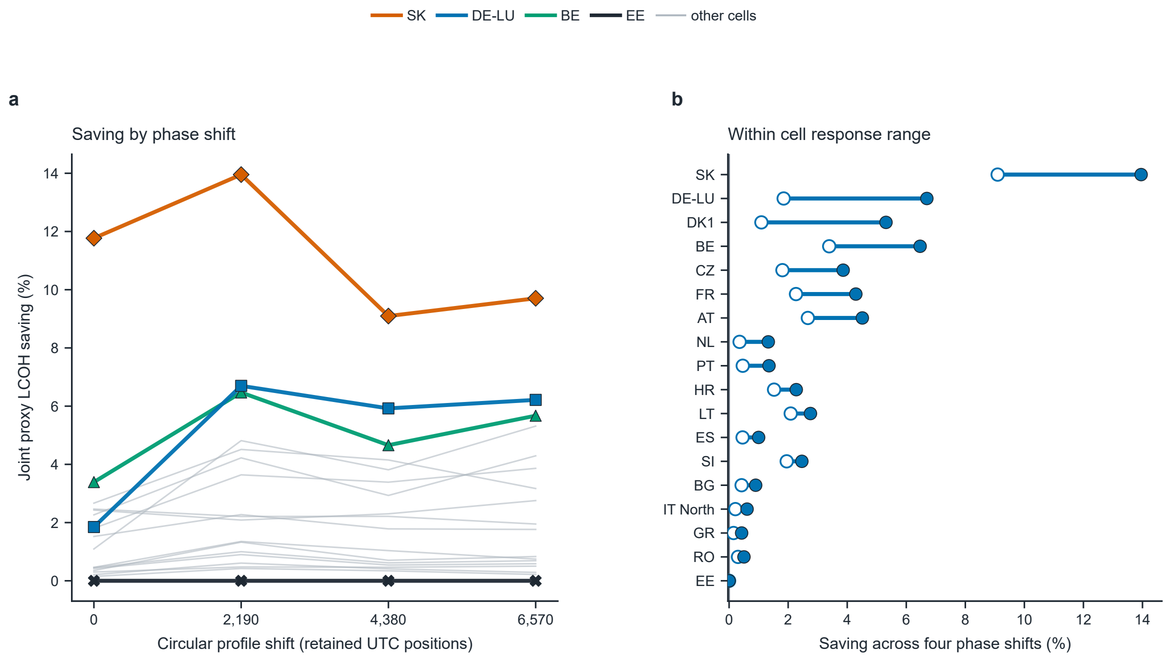
Cell	LCOH (EUR/kg; renewable only -> proxy)	Saving (%)	Wind change (MW)	Solar change (MW)	Electrolyzer change (MW)	Model qualified grid (MWh)
SK	5.279 -> 4.657	11.77	0.0	-58.8	-20.6	74,547
BE	4.192 -> 4.049	3.39	-13.1	-0.7	3.8	23,087
AT	3.657 -> 3.559	2.67	0.2	-16.7	2.9	21,165
SI	4.243 -> 4.138	2.46	-9.8	-1.0	0.7	16,364
LT	3.416 -> 3.333	2.43	-8.1	0.0	-0.5	15,604
FR	3.408 -> 3.331	2.27	-10.0	-0.2	2.0	17,844
DE-LU	3.645 -> 3.578	1.85	-4.6	-1.8	2.9	23,835
CZ	3.598 -> 3.533	1.81	-1.8	-9.6	2.8	16,126
HR	3.431 -> 3.379	1.52	-2.1	-4.3	0.0	11,741
DK1 (country profile)	3.569 -> 3.530	1.09	-4.4	0.0	1.8	17,885
PT	3.117 -> 3.102	0.46	-2.1	0.6	0.3	2,919
ES	2.882 -> 2.869	0.46	-2.7	1.4	0.5	2,938
BG	3.875 -> 3.859	0.42	-2.1	0.3	0.3	2,725
NL	4.201 -> 4.187	0.35	-1.5	-0.4	0.2	2,549
RO	3.725 -> 3.714	0.30	-1.4	0.3	0.2	1,796
IT North (country profile)	3.171 -> 3.165	0.21	-0.9	0.1	0.0	1,449
GR	3.067 -> 3.063	0.15	0.3	-1.4	0.3	901
EE	1.887 -> 1.887	0.00	0.0	0.0	0.0	0

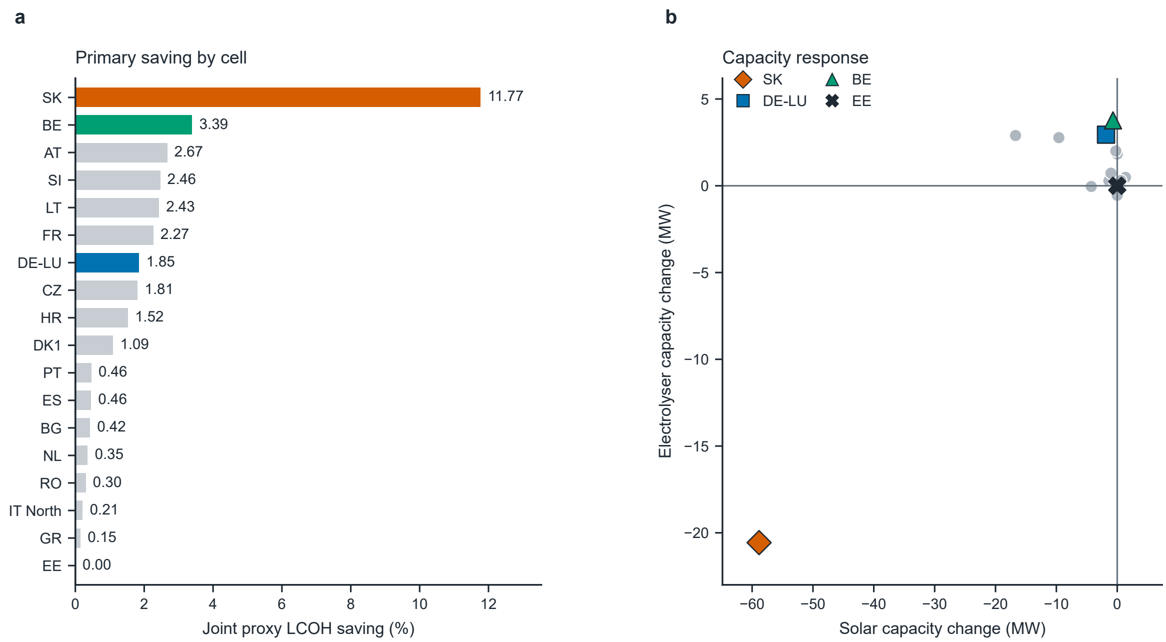
Table 4. Cross period and published profile sensitivity.

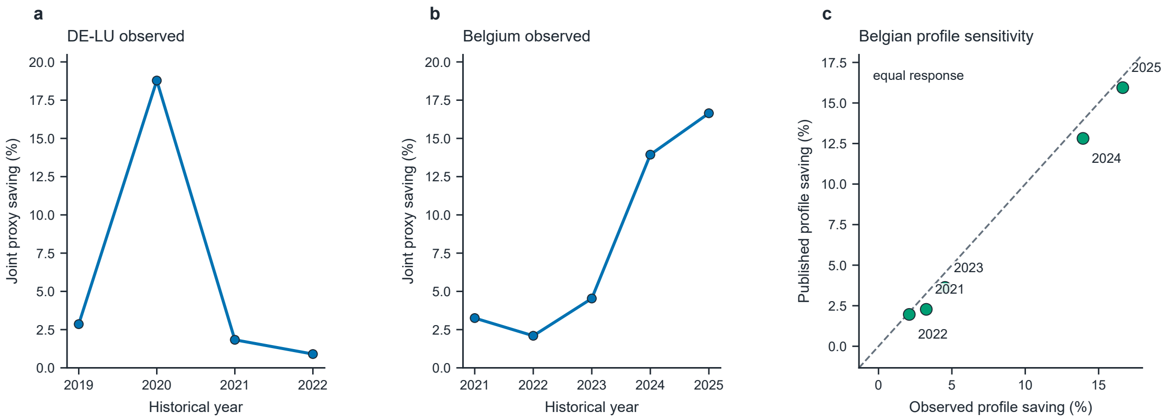
Area	Year	Profile	Joint saving (%)
BE	2021	dayahead_forecast	2.28
BE	2021	observed_generation	3.25
BE	2022	dayahead_forecast	1.96
BE	2022	observed_generation	2.10
BE	2023	dayahead_forecast	3.63
BE	2023	observed_generation	4.54
BE	2024	dayahead_forecast	12.81
BE	2024	observed_generation	13.94
BE	2025	dayahead_forecast	15.95
BE	2025	observed_generation	16.65
DE-LU	2019	observed_generation	2.86
DE-LU	2020	observed_generation	18.78
DE-LU	2021	observed_generation	1.83
DE-LU	2022	observed_generation	0.90

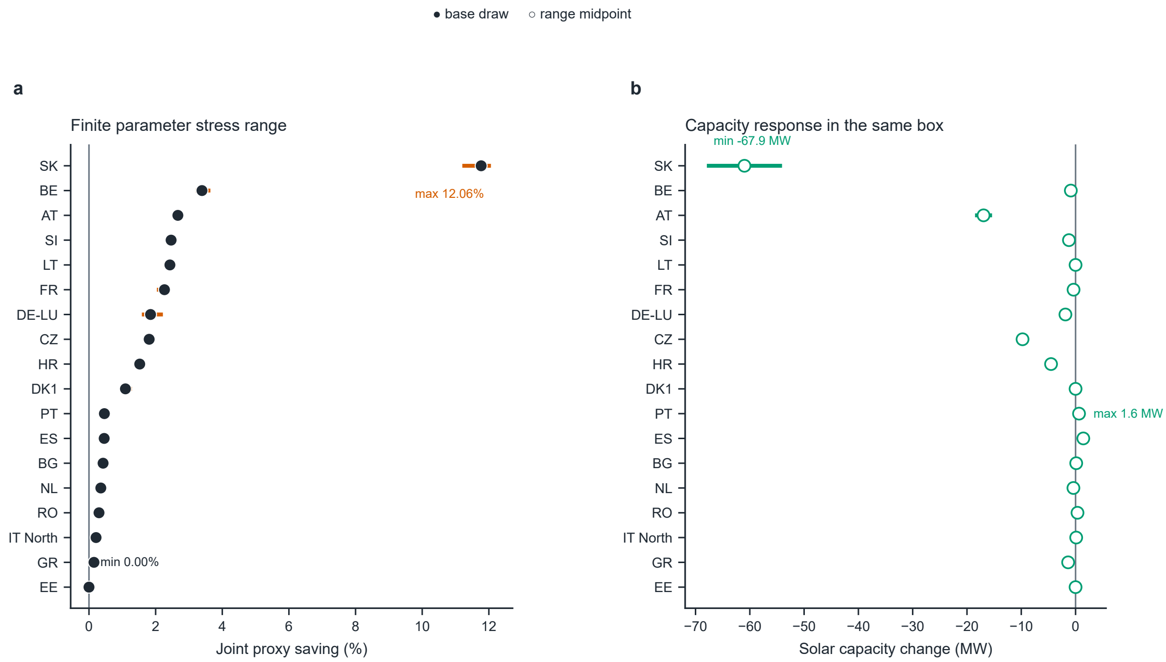
Table 5. Bounded parameter stress results.

Cell	Base saving (%)	Minimum (%)	Maximum (%)	Range (pp)	Zero response draws	Positive response draws
AT	2.67	2.49	2.78	0.30	0	13
BE	3.39	3.20	3.64	0.45	0	13
BG	0.42	0.39	0.44	0.05	0	13
CZ	1.81	1.66	1.90	0.23	0	13
DE-LU	1.85	1.59	2.22	0.63	0	13
DK1 (country profile)	1.09	0.97	1.29	0.32	0	13
EE	0.00	0.00	0.00	0.00	12	1
ES	0.46	0.42	0.48	0.06	0	13
FR	2.27	2.04	2.38	0.34	0	13
GR	0.15	0.14	0.16	0.02	0	13
HR	1.52	1.40	1.59	0.19	0	13
IT North (country profile)	0.21	0.19	0.22	0.03	0	13
LT	2.43	2.24	2.52	0.29	0	13
NL	0.35	0.33	0.37	0.04	0	13
PT	0.46	0.43	0.48	0.06	0	13
RO	0.30	0.28	0.31	0.03	0	13
SI	2.46	2.33	2.54	0.22	0	13
SK	11.77	11.20	12.06	0.86	0	13


SK
DE-LU
BE
EE
other cells
a
Saving by phase shift
Joint proxy LCOH saving (%)
Circular profile shift (retained UTC positions)
0
2,190
4,380
6,570
b
Within cell response range
SK
DE-LU
DK1
BE
CZ
FR
AT
NL
PT
HR
LT
ES
SI
BG
IT North
GR
RO
EE
Saving across four phase shifts (%)


a
Primary saving by cell
SK
BE
AT
SI
LT
FR
DE-LU
CZ
HR
DK1
PT
ES
BG
NL
RO
IT North
GR
EE
11.77
3.39
2.67
2.46
2.43
2.27
1.85
1.81
1.52
1.09
0.46
0.46
0.42
0.35
0.30
0.21
0.15
0.00
0
2
4
6
8
10
12
Joint proxy LCOH saving (%)
b
Capacity response
SK
BE
DE-LU
EE
5
0
−5
−10
−15
−20
Electrolyser capacity change (MW)
−60
−50
−40
−30
−20
−10
0
Solar capacity change (MW)


a
DE-LU observed
Joint proxy saving (%)
Historical year
b
Belgium observed
Joint proxy saving (%)
Historical year
c
Belgian profile sensitivity
equal response
Published profile saving (%)
Observed profile saving (%)
2025
2024
2023
2021
2022



• base draw ○ range midpoint
a
Finite parameter stress range
SK
BE
AT
SI
LT
FR
DE-LU
CZ
HR
DK1
PT
ES
BG
NL
RO
IT North
GR
EE
max 12.06%
min 0.00%
0
2
4
6
8
10
12
Joint proxy saving (%)
b
Capacity response in the same box
min -67.9 MW
max 1.6 MW
−70
−60
−50
−40
−30
−20
−10
0
Solar capacity change (MW)

Figure captions

Figure 1. Temporal coincidence mechanism (E1). Prices and trigger counts are fixed while renewable profiles rotate; the within cell saving range has a median of 0.82 percentage points and reaches 4.86 points, showing the role of temporal alignment within the model. The zero shift point reproduces the primary screen reported in Table 3. Open and filled endpoints in panel b denote the minimum and maximum phase shift responses.

Figure 2. Primary screen: LCOH savings and capacity substitutions across 18 selected cells. SK reaches 11.77% saving with -58.8 MW solar and -20.6 MW electrolyzer capacity; EE shows zero response under the declared model boundary.

Figure 3. Cross period sensitivity (E2). Joint savings range from 0.90% to 18.78% across 14 retained source year/profile cells; branch attribution varies by year.

Figure 4. Bounded parameter stress (E3). The finite engineering box yields full minimum to maximum response ranges from 0.00% to 12.06% across the selected cells; the global saving endpoints are labeled in panel a and detailed cell endpoints are reported in Table 5. Twelve zero response joint comparisons are retained alongside positive responses. The ranges are engineering bounds, not probability intervals.